

Comprehensive Comparisons of Grafting-Modified Different Polypropylene as HVDC Cable Insulation Material

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Abstract—Polypropylene (PP) is considered to be a rather potential material for recyclable HVDC cable insulation. Grafting modification has been proven to be effective to further enhance the dielectric properties of PP. To investigate the grafting effect on the performance of different PPs, homopolymerized PP (PPH), ethylene-propylene copolymer (PEC), and impact PP copolymer (IPC) are, respectively, grafting-modified by styrene, and the comprehensive properties are experimentally studied. The results indicate that all these grafting-modified PPs show enhanced dielectric properties, while the enhancement levels differ. Styrene-grafted PPH presents the best dielectric properties improvement and highest thermostability, but the elastic modulus is also the highest. PEC exhibits moderate dielectric properties enhancement after grafting modification. Grafting-modified IPC shows excellent mechanical properties, but the enhancements on dielectric properties are rather limited. Grafting styrene has no significant influence on the thermal and mechanical properties of PP. In conclusion, PPH is preferred to be the matrix for styrene-grafting to achieve enhanced dielectric properties. However, the mechanical properties still need further modification to meet the demands of cable insulation.

Index Terms—Dielectric properties, ethylene-propylene copolymer (PEC), grafting modification, impact polypropylene (PP) copolymer (IPC), polypropylene (PP), styrene.

I. INTRODUCTION

DEVELOPMENT and production of the insulation material is a key technique of HVDC cable, even the bulk power HVDC transmission system [1], [2], [3], [4], [5]. So far, cross-linked polyethylene (XLPE) is extensively utilized as the insulation material of cable, with its merits of high dielectric performance, excellent mechanical properties, and low cost

[6], [7]. However, with the continuously growing transmission capacity of power cables, XLPE (highest operation temperature about 90 °C) is no longer able to meet the higher demand of operating temperature. In addition, cross-linking process makes XLPE a thermoset material, thus unrecyclable after lifetime expired. Traditional disposing way, incineration, will severely pollute the environment [8]. Therefore, it is necessary to find a new highly thermoresistant, recyclable insulation material to replace XLPE [9].

Polypropylene (PP) is a competitive and potential material for HVDC cable insulation. Compared with XLPE, PP shows better thermostability and superior dielectric performance, especially under high temperature [10]. In addition, PP is a main product of petrochemical industry with lower cost, and the thermoplastic nature of PP makes it easier to be recycled and reprocessed [11]. All of these advantages indicate that PP is a promising material for recyclable HVDC cable insulation.

Grafting modification is an effective method to further enhance the dielectric properties of PP for extra-high and ultrahigh DC power cable [12], [13], [14]. By grafting chemical groups onto PP molecular chains, PP is microscopically modified and the macroscopic dielectric properties can be improved. Yuan *et al.* [15], [16] found that grafting-modifying PP by styrene can introduce deep traps in the PP matrix, and the crystallization of PP is simultaneously influenced. As a result, the dielectric properties including DC breakdown strength and DC volume resistivity are significantly enhanced especially under high temperature, and the space charge accumulation is suppressed. However, the PP matrix used for grafting modification in aforementioned works is isotactic PP, of a simple, regular microstructure, while the generalized “PP” includes a great diversity of sorts, such as copolymerized PP, impact PP copolymer (IPC), and so on. The different physical composition and chemical structures of PP may have effect on the performance of grafting modification, which needs further investigation.

In this article, three kinds of PP—homopolymerized PP (PPH), ethylene-propylene copolymer (PEC), and IPC, are chosen as the PP matrices and grafting-modified by styrene, respectively. The thermal, mechanical, and dielectric properties are experimentally investigated and analyzed. A comprehensive evaluation and comparison of them are performed to pick out the best PP matrix for grafting-modified PP-based HVDC cable insulation material.

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TABLE I
 M_w AND COMPOSITION OF PP MATRICES

Designation	M_w (g/mol)	Ethylene content (%)
PPH	387000	0
PEC	367000	3.2
IPC	454000	20.1

II. EXPERIMENTAL PROCEDURE

A. Materials

Three types of PP are all provided by SINOPEC. PPH is the isotactic PP homopolymerized by liquid-phase propylene. PEC is randomly copolymerized by liquid phase propylene and gaseous phase ethylene. IPC consists of an PEC (copolymerized in gaseous phase) and a rubbery phase of high ethylene content. The weight-mean molecular weight (M_w) and ethylene content of these PP matrices are shown in Table I.

Styrene (St) is chosen as the grafting monomer, and benzoyl peroxide (BPO) is used as an initiator. Both of them are provided by Beijing J&K Technology Co. Ltd. Before reaction, styrene is purified by vacuum distillation to prevent self-polymerization. The grafting content is optimized as 5 phr according to the related research [16]. Here, phr means the gram weight of St per 100 g of the PP matrix.

B. Grafting Reaction and Preparation

The grafting process is performed in a high-pressure steel reactor. Pure nitrogen is adopted to purge residual air in the reactor. St of 5 phr and BPO with moderate content are mixed evenly into uniform solution and then added into the reactor with PP matrix. During the grafting reaction, the mixture is stirred for 0.5 h and then heated to 90 °C for another 4 h. After the reaction accomplished, the reactor is repurged by pure nitrogen, and the reacted material is filtered and dried. According to the PP type used in the reaction, the grafting-modified PP materials are, respectively, named as PPH-g-St, PEC-g-St, and IPC-g-St.

These prepared grafting-modified materials are first compression molded at 200 °C under the pressure of 10 MPa for 15 min and then water-cooled to 25 °C under 15 MPa for 5 min. Thus, film samples of grafting-modified PP are prepared for subsequent characterizations. The unreacted PP matrices, i.e., PPH, PEC, and IPC, are taken the same compression procedure as references. All the prepared film samples present decent surface roughness and uniform thickness.

C. Characterizations

Fourier-transform infrared (FT-IR) spectrometry (Thermo Fisher Nicolet iS10, USA) is carried out to characterize the grafted styrene group. Transmission mode and the resolution of 4 cm^{-1} is set during the experiment. The crystallization of grafting-modified PP is investigated by X-ray diffraction (XRD, Philips X' Pert MPD, The Netherlands). The 2θ angle scanning range is 5°–30°. Polarizing microscope (POM, Zeiss Axio Imager A1m, Germany) is adopted to observe the crystallization morphology. The POM samples are heated to

200 °C for 3 min to eliminate heat history and then cooled to 25 °C and the POM images are taken.

Differential scanning calorimetry (DSC, TA Q2000, USA) is used to investigate the thermal properties. The samples are first heated to 200 °C for 5 min to erase heat history and then rapidly cooled to 10 °C by 10 °C/min to measure the heat flow of crystallization process. After that, the samples are heated again to 200 °C to measure the melting process. A tensile tester (MTS CMT4304, USA) is used to investigate the mechanical properties of the samples. The samples are of dumbbell shape according to the test standard ASTM D638-2003. The tensile speed is controlled as 20 mm/min. Five specimens for each sample are tested to ensure the repeatability.

DC leakage current of the samples is measured by a three-electrode platform and an electrometer (Keithley 2635B, USA). To investigate the high-temperature performance, the measurement is performed under 30 °C, 90 °C, and 110 °C. The samples are of 100 μm thickness and sputtering-coated with gold before measurement to ensure the sufficient contact with the electrodes. The applied electric field rises from 5 to 80 kV/mm by the step of 5 kV/mm. In each measurement period, the stabilized leakage current is recorded and used to calculate DC volume conductivity according to IEC Standard 60093. The space charge accumulation is tested by pulsed electroacoustic (PEA) method at 30 °C. The 200- μm -thickness film sample is polarized by a 50-kV/mm DC electric field for 30 min. Meanwhile, the PEA waveform is shown on the screen of an oscilloscope and recorded by computer. A DC high voltage tester (Haiwo High Voltage, China) is used to test the DC breakdown strength of samples. The sample is of 100 μm mean thickness and sandwiched by sphere-plate electrodes bathing in silicone oil (preventing surface discharge). The same to the DC volume conductivity measurement, the breakdown strength test is also performed under the three temperatures as above. A 1-kV/s-rising DC voltage is applied between the electrodes until breakdown. The breakdown strength E is calculated by $E = U/d$, where U is the tested breakdown voltage and d is the sample's thickness. For each sample, 20 breakdown points are tested, and the corresponding breakdown characteristics are analyzed by two-parameter Weibull distribution.

III. EXPERIMENTAL RESULTS

A. Characterization of Grafted Group

Fig. 1 exhibits the FT-IR spectra of the three types of PP, pristine, and grafting-modified. After grafting styrene, as the arrows in Fig. 1 guide, one can observe five emerging absorption peaks, which correspond to the vibration of aromatic ring: 698, 758, 1500, 1601, and 3050–3130 cm^{-1} (an absorption peak group consists of three small peaks) [17]. Thus, it can be confirmed that styrene groups are successfully grafted onto PP molecular chain.

B. Crystallization

XRD is adopted to investigate the crystallization form and crystallinity. The results are shown in Fig. 2 and Table II. The diffraction peaks located at 14.1°, 16.9°, 18.5°, 21.1°, and

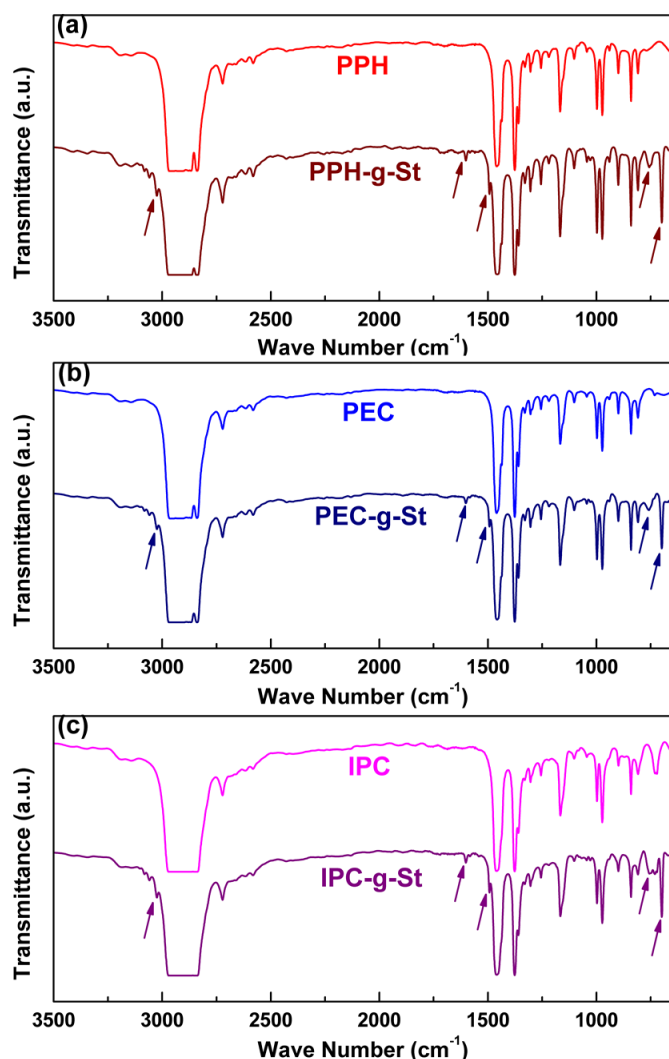


Fig. 1. FT-IR spectra of three types of PP before and after grafting styrene. (a) PPH and PPH-g-St. (b) PEC and PEC-g-St. (c) IPC and IPC-g-St.

TABLE II
CRYSTALLINITY OF THREE TYPES OF PP BEFORE AND AFTER GRAFTING STYRENE

Designation	PPH	PPH-g-St	PEC	PEC-g-St	IPC	IPC-g-St
Crystallinity	56.2%	50.4%	52.5%	46.5%	27.8%	25.0%

21.9° correspond to the crystal faces of (110), (040), (130), (111), and (131), which belong to the alpha crystal form of PP [18]. This result indicates that the six materials are all of alpha crystal form. The crystallinity of these materials shown in Table II is calculated by the area ratio of crystallization peaks and amorphous phase. All the grafting-modified PP shows a lower crystallinity than the pristine one. Among them, IPC with the highest ethylene content shows the lowest crystallinity, while PPH, which is isotactic and of highly regular structure, presents the highest crystallinity.

Crystallization morphology of the materials is observed by POM microscopy, as Fig. 3 shows. Both untreated PPH

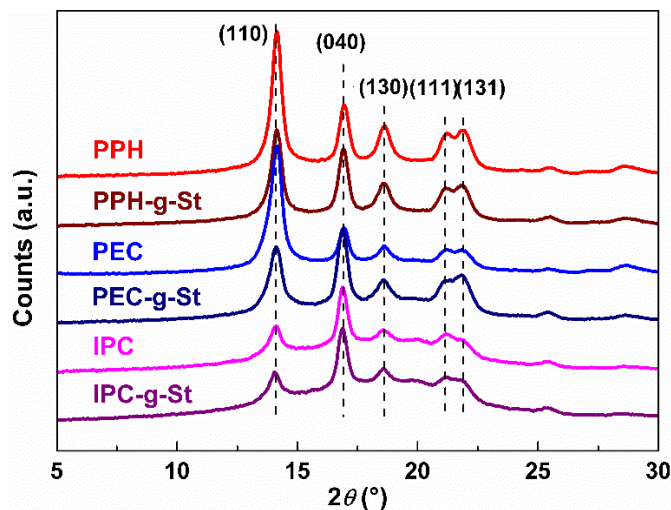


Fig. 2. XRD spectra of three types of PP before and after grafting styrene.

TABLE III
THERMAL PARAMETERS OF THREE TYPES OF PP BEFORE AND AFTER GRAFTING STYRENE

Designation	PPH	PPH-g-St	PEC	PEC-g-St	IPC	IPC-g-St
Melting temp.(°C)	158.2	154.1	136.2	137.2	126.0	122.3
Melting peak temp. (°C)	161.0	159.0	142.9	144.1	138.4	137.9
Crystallization temp. (°C)	118.0	117.9	103.0	107.5	105.0	103.0
Melting enthalpy (J/g)	103.1	98.4	73.8	71.4	31.6	30.9

and PEC show large spherulite size and clear boundaries of crystallization region, whereas the spherulites in IPC are too tiny to observe. The field of vision is also dark due to the low crystallinity.

After grafting modification, the spherulite sizes of PPH-g-St and PEC-g-St are reduced, with fuzzified boundaries. Compared with PPH-g-St, the crystallization in PEC-g-St is of less regularity, more indistinct boundaries, and imperfect spherulites. This may be the reason why the crystallinity of PEC-g-St is lower than that of PPH-g-St. Regarding IPC, it can be observed that the crystalline grain is a little bit more distinct after grafting modification, however, still no significant improvement on the fuzzy crystallization. The crystallinity is still at the lowest level.

C. Thermal Properties

The heat flow curves during crystallization and melting process and corresponding thermal parameters obtained by DSC measurement are shown in Fig. 4 and Table III. Concerning untreated samples, nearly all thermal parameters show the trend of PPH > PEC > IPC, which is opposite to the ethylene content exhibited in Table I. After grafting modification, the aforementioned trend remains unchanged. The enthalpies of the three materials all decrease, which coincides with the crystallinity shown in Table II. Regarding PPH and PEC, both the crystallization temperature and melting temperature

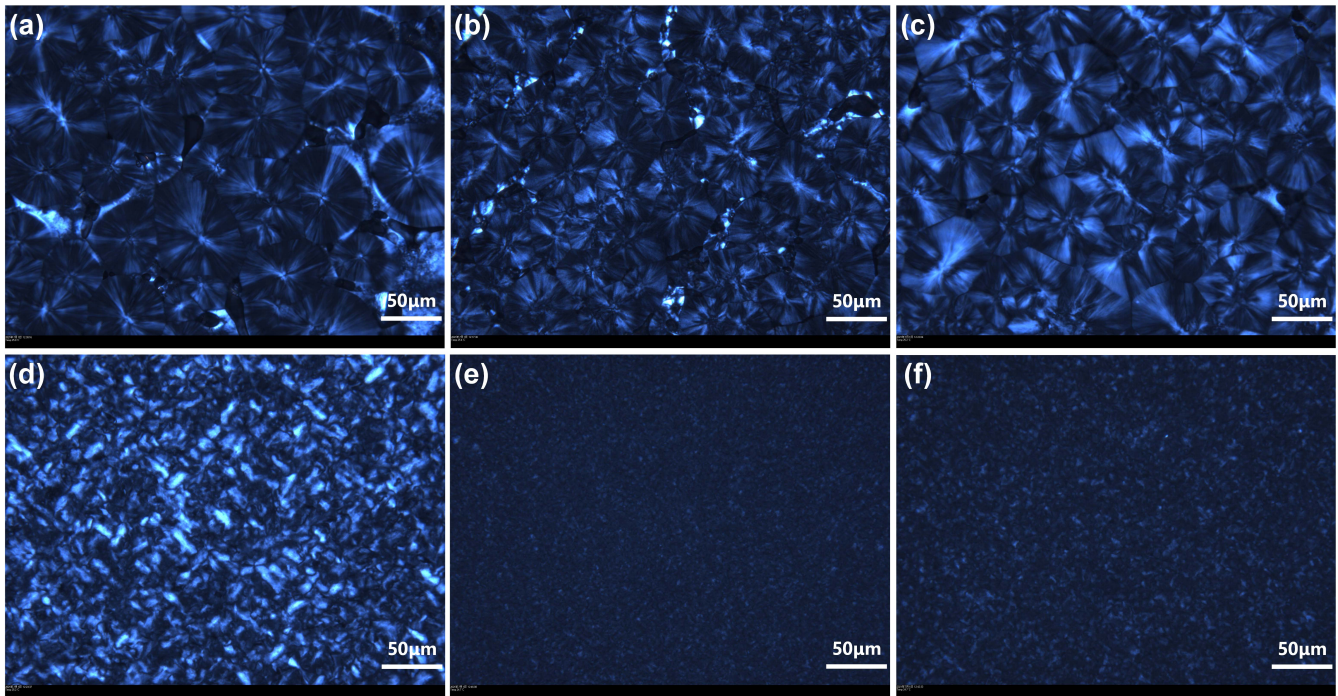


Fig. 3. POM images of three types of PP before and after grafting styrene. (a) PPH. (b) PPH-g-St. (c) PEC. (d) PEC-g-St. (e) IPC. (f) IPC-g-St.

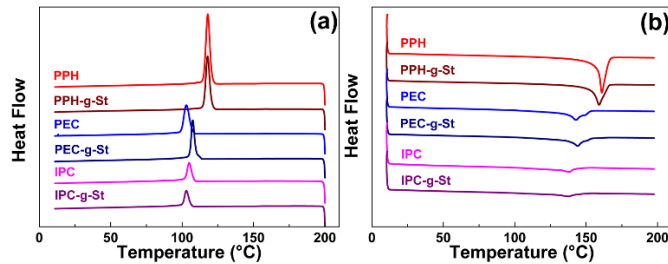


Fig. 4. DSC curves of three types of PP before and after grafting modification. (a) Crystallization process. (b) Melting process.

TABLE IV

MECHANICAL PROPERTIES OF THREE TYPES OF PP BEFORE AND AFTER MODIFICATION

Designation	PPH	PPH-g-St	PEC	PEC-g-St	IPC	IPC-g-St
Elastic modulus (MPa)	636 ±52	643 ±17	449 ±7	351 ±16	83 ±15	52 ±7
Elongation-at-break (%)	1398 ±83	1202 ±78	1447 ±208	1347 ±184	1663 ±109	1620 ±185
Tensile strength (MPa)	52±5	46±4	45±3	40±3	25±1	25±3

are reduced after grafting modification, whereas in PEC-g-St, these two parameters show slight increase than PEC.

D. Mechanical Properties

The mechanical properties of the pristine PP and modified PP tested by the tensile tester are shown in Fig. 5 (stress-strain curve) and Table IV (mechanical parameters). It can be observed that the three mechanical indices all decrease after grafting modification regardless of PP type (except for the elastic modulus of PPH, presenting a slight increase which

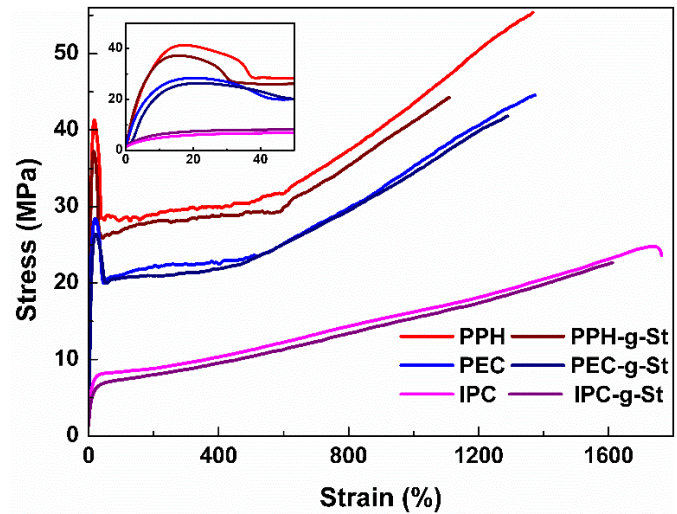


Fig. 5. Stress-strain curves of three types of PP before and after grafting styrene.

may be due to the random error). Among them, the elongation-at-break and tensile strength are all very high, which outstrip the related standard requirement for XLPE (elongation-at-break > 200% and tensile strength > 12.5 MPa) [19].

Compared among the three types of PP, the elastic modulus and tensile strength display the trend of PPH > PEC > IPC, while the elongation-at-break is opposite. Grafting modification does not change this tendency. Interestingly, the variation tendency of these mechanical properties coincides with the crystallinity exhibited in Section III-B. High crystallinity usually means that the polymer is stiff and of high elastic modulus, high tensile strength, and low elongation-at-break. With the most regular molecular structure, PPH shows the highest crystallinity, and the elastic modulus and tensile strength are also the highest, while regarding IPC, the PP-based copolymer with

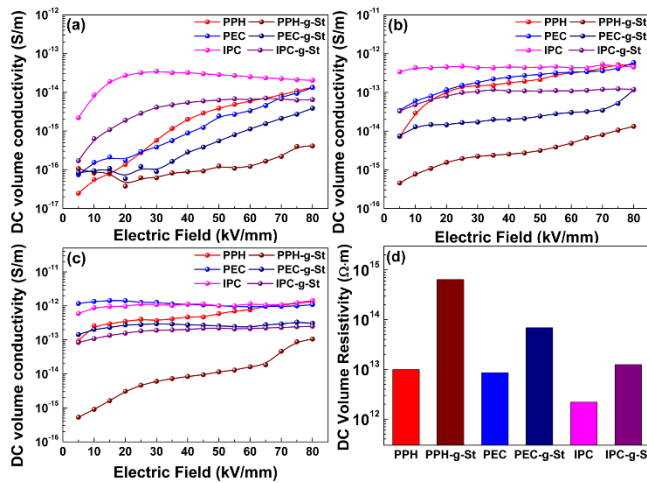


Fig. 6. DC volume conductivity of the three types of PP before and after grafting modification under (a) 30 °C, (b) 90 °C, (c) 110 °C, and (d) DC volume resistivity under 20 kV/mm, 90 °C.

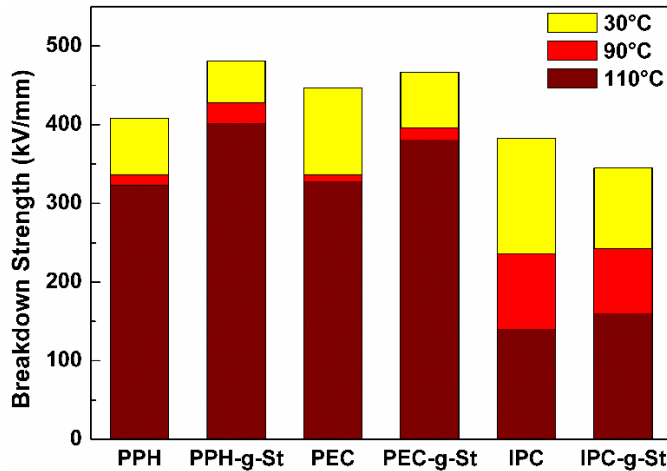


Fig. 7. Characteristic breakdown strength of three types of PP before and after grafting modification under 30 °C, 90 °C, and 110 °C.

impact resistance and complex structure, the crystallinity is the lowest. So that IPC shows the highest elongation-at-break, and the other two parameters are the lowest. For the effect of grafting, the grafted styrene group on PP molecular chain can further hinder the crystallization, due to the steric effect, eventually leading to lower crystallinity. This would be the main reason why grafting-modified PPs show reduced elastic modulus and tensile strength than corresponding pristine one.

E. DC Volume Conductivity

DC volume conductivity, which is calculated from DC leakage current, can be used to evaluate the insulating performance of the material directly. Lower conductivity and higher resistivity usually mean lower energy loss and better insulating performance. The DC volume conductivity of the three types of PP before and after grafting modification under various electric field and temperature is shown in Fig. 6(a)–(c). Considering that the highest electric field of insulation layer in HVDC cable is not more than 20 kV/mm [20] and the designed operation temperature of PP-based

TABLE V
DC BREAKDOWN PARAMETERS OF TEST SAMPLES

Designation	30°C		90°C		110°C	
	E_c (kV/mm)	β	E_c (kV/mm)	β	E_c (kV/mm)	β
PPH	408	6.7	336	14.0	323	10.9
PPH-g-St	481	9.4	428	9.4	401	11.0
PEC	447	12.9	336	16.3	327	12.6
PEC-g-St	467	10.7	396	10.7	380	14.1
IPC	383	13.6	236	28.7	140	23.6
IPC-g-St	345	9.0	242	34.8	160	18.8

insulation is 90 °C [21], the DC volume resistivity values under this condition are calculated and shown in Fig. 6(d).

It is easy to observe that after grafting modification, the DC volume conductivities of the three types are all significantly suppressed. The relationship among pristine materials is $PPH > PEC > IPC$ and so do the materials after grafting modification. Under operation condition, as Fig. 6(d) shows, the DC volume resistivity of PPH-g-St exhibits the enhancement of approximately two orders of magnitude, which is the most significant and highest than other materials. The significant suppression of conductivity by grafting styrene can be attributed to the introduced large quantities of deep traps [15]. These results indicate that grafting modification is effective to enhance the DC volume resistivity regardless of the PP type. Even under a higher temperature of 110 °C, the grafting-modified PPs still present fairly low conductivity, especially for PPH-g-St.

F. DC Breakdown Strength

To evaluate the DC voltage toleration level of these materials, DC breakdown strength is tested and analyzed by Weibull distribution

$$P(E) = 1 - \exp(-(E/E_c)^\beta) \quad (1)$$

where $P(E)$ is the accumulative breakdown probability, E is the breakdown strength experimental value, E_c is the characteristic breakdown strength corresponding to the accumulative breakdown probability of 63.2%, and β is the shape parameter of Weibull fitting. The analyzed E_c and β are shown in Table V and Fig. 7. The detailed Weibull fitting plots can be found in Appendix.

The test results above indicate that grafting modification can effectively enhance the DC breakdown strength of PPH and PEC under three temperatures. Particularly, PPH-g-St can still maintain a fairly high strength over 400 kV/mm under 110 °C, while in the case of IPC, the improvement of grafting modification is very little. E_c of IPC under 30 °C even decreases after grafting modification. Still, the trend of breakdown strength is $PPH > PEC > IPC$. Therefore, it can be concluded that the grafting modification by styrene is only effective to enhance the DC breakdown performance of PPH and PEC, whereas almost in vain for IPC.

G. Space Charge Behavior

The dynamic space charge distributions measured by PEA method under 30 °C are exhibited in Fig. 8. Poisson equation (2) is employed to calculate the electric field distortion

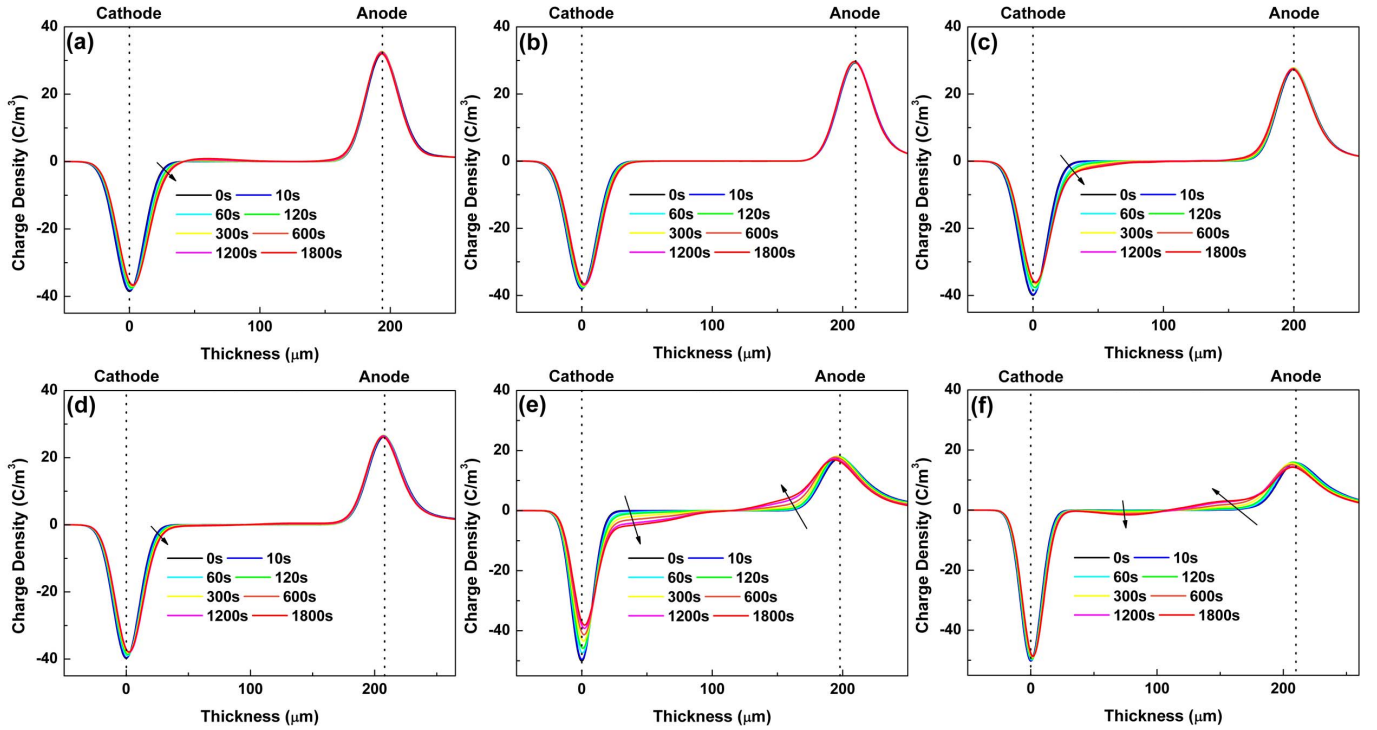


Fig. 8. Dynamic space charge distribution of three types of PP before and after grafting modification. (a) PPH. (b) PPH-g-St. (c) PEC. (d) PEC-g-St. (e) IPC. (f) IPC-g-St.

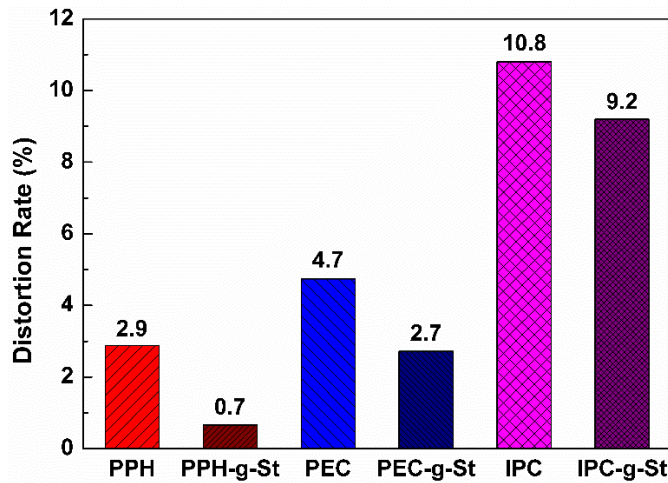


Fig. 9. Electric field distortion rate of three types of PP before and after grafting modification.

triggered by space charge accumulation

$$E(x) = \frac{1}{\epsilon_0 \epsilon_r} \int_0^d \rho(x) dx \quad (2)$$

where ϵ_0 and ϵ_r are the vacuum permittivity and relative permittivity of PP, respectively. d is the sample thickness. $\rho(x)$ and $E(x)$ are the space charge density and electric field distribution, respectively. The electric field distortion rate ΔE is calculated by the following equation:

$$\Delta E = (E_{\max} - E_{\text{av}}) / E_{\text{av}} \times 100\% \quad (3)$$

where E_{av} equals 50 kV/mm as mentioned in Section II-C. $E_{\max}$ is the highest electric field throughout the polarization. The calculation results are shown in Fig. 9.

One can find that all the PP materials show suppressed space charge accumulation after grafting styrene, and the electric field distortion rates are also reduced. The improvement of space charge accumulation is significant in the case of PPH and PEC. Especially in PPH-g-St, the space charge injection can hardly be seen, and the distortion rate is of the lowest level. However, regarding IPC, highly injected and accumulated space charge are observed near both electrodes, as the guidance arrows indicate. Grafting modification can suppress the injection of IPC in a certain extent but cannot improve the accumulation fundamentally, so that the improvement of distortion rate is limited, not as significant as PPH and PEC. The PEA test results indicate that grafting styrene can achieve effective space charge suppression in PPH and PEC, in which PPH-g-St performs the best.

IV. DISCUSSION

As these experimental results show, the thermal properties, mechanical properties, and dielectric properties of PPs are all affected by grafting styrene, with various effects with respect to different PP types. Generally, grafting styrene has very limited influence on the thermal and mechanical properties of PP, while significant on the dielectric properties. Among the three types of PP, PPH with the most regular structure shows the best dielectric properties enhancement. PEC is moderate, and IPC is at the lowest level.

Considering the strictly identical grafting procedure and sample preparation, the property difference is strongly correlated with the inherent nature of PP matrix. As Section III-B shows, the crystallinity of three types of PP is as $\text{PPH} > \text{PEC} > \text{IPC}$ and further decreases after grafting styrene. This can be explained as that the random copolymer of ethylene and

propylene in PEC and IPC is difficult to shape into regular arrangements, so that the crystallization is not perfect as PPH and resulting in low crystallinity. A negative correlation can be observed between crystallinity and ethylene content. In addition, the grafted aromatic ring with high rigidity would have steric effect on the motion of PP molecular chain and hinder the regular arrangements of molecules, finally resulting in further lower crystallinity. The crystallization morphology affected by grafting modification can be directly compared in the POM images of Fig. 3. Both PPH and PEC show large spherulites and clear boundaries, where the spherulite size of PPH is a little larger than that of PEC. After grafting modification, as Fig. 3(b) and (d) shows, the spherulite sizes of PPH-g-St and PEC-g-St are reduced and the boundaries become fuzzy. Among them, PEC-g-St shows imperfect spherulites and the boundaries is more indistinct. This would be due to the combined effect of random copolymerized molecular chain and the grafted aromatic group that hinder the regular molecular arrangements and crystallization. Regarding IPC, the pristine material is of complicated structure and presents extremely tiny spherulites. Thus, although grafted styrene group is introduced in IPC-g-St, the crystallization morphology is not significantly influenced.

The thermal parameters trend of $\text{PPH} > \text{PEC} > \text{IPC}$ can be explained by the regularity of molecular chains. High molecule regularity in PPH means compact molecular chain arrangements and strong Van der Waals force, thus exhibiting higher melting temperature and easier crystallization (high crystallization temperature). The grafted styrene in modified material would destroy the regularity, thus lowering the thermal parameters compared with pristine one. However, in the case of PEC-g-St, the highly fuzzified and complicated spherulite boundaries [as Fig. 3(d) shows] can enhance the affinity among the spherulites and further hinder the molecular motion. Thus, melting and crystallization temperature are slightly raised compared with pristine PEC. Generally speaking, however, due to low grafting content of styrene (about 5%), the changes in thermal properties are limited.

The mechanical properties of these material are also correlated with the crystallinity, as discussed in Section III-D. After grafting modification, the crystallinity decreases and results in low elastic modulus and tensile strength. However, the elongation-at-break is also reduced. This can be explained as that the flexibility of grafting-modified molecular chain decreases due to the high rigidity and steric effect of grafted aromatic groups. In summary, like thermal properties, the grafting modification effect on mechanical properties is also of small changes. The modified materials can meet the elongation-at-break and tensile strength demands of IEC standard, but as cable insulation, the elastic modulus of PPH and PEC is still too high (over 300 MPa) compared with XLPE (around 200 MPa [22]), which would need further modification.

Section III-E to III-G exhibits the various dielectric properties improvements by grafting styrene with respect to different types of PP. Among them, PPH-g-St performs the best. For DC volume conductivity, all PPs shows significant enhancements under three temperatures, which can be attributed to the deep

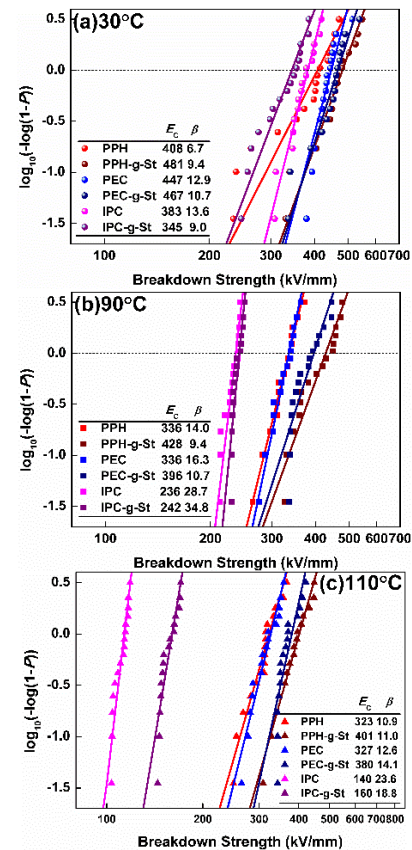


Fig. 10. DC breakdown Weibull plots of three types of PP before and after grafting modification.

traps introduced by grafted groups [15], whereas regarding the other two dielectric properties, the results need more detailed analysis.

Regarding DC breakdown strength, PPH-g-St and PEC-g-St show significant enhancements especially under high temperature, while the enhancement is little in IPC-g-St. This could be explained by the crystallization morphology, as shown in Fig. 3. Spherulite boundaries and amorphous areas are usually of high charge carrier conductivity and can act as transmission path [16]. After grafting modification, the spherulite size is reduced and the boundaries are indistinct in PPH-g-St and PEC-g-St. This would result in more spherulite boundaries and longer charge transmission route. Therefore, the charge transmission path of DC breakdown process is lengthened and the strength is raised. However, the fuzzified boundaries also come along with reduced crystallinity and more amorphous areas, which are negative to the dielectric properties. This would be a possible reason why PEC-g-St is not as well as PPH-g-St on DC conductivity and breakdown strength. For IPC, whose crystallinity is the lowest, the crystallization morphology shows little improvement after grafting modification, so that these two dielectric properties of IPC-g-St are enhanced just in a limited range, inferior to the PPH and PEC.

Space charge accumulation results show a similar trend to the other two dielectric properties. The deep traps introduced by grafted groups are responsible for the significant suppression of space charge injection and accumulation [13], [15], but for IPC-g-St, though one can observe the improvement of injection after modification, the accumulation is still severe. IPC with low crystallinity is of complicated structure including

ethylene-propylene copolymerized phase and rubbery phase. Large quantities of physical defects abound in the interface of them and amorphous areas, which can introduce abundant shallow traps. These shallow traps can capture charges, thus leading to space charge accumulation [23]. Therefore, the space charge accumulation in IPC deteriorates, finally resulting in high electric field distortion rate.

V. CONCLUSION

In this article, PPH, PEC, and IPC are modified by grafting styrene. Comprehensive performances including thermal, mechanical, and dielectric properties are experimentally investigated and compared. The conclusions are listed as follows.

- 1) Grafting styrene does not change the crystallization form but reduces the crystallinity. Crystallinity: PPH > PEC > IPC, so does the grafting-modified PPs. IPC shows no obvious change in crystallization morphology after grafting modification, while PPH-g-St and PEC-g-St exhibit reduced spherulite size and indistinct spherulite boundaries. In addition, imperfect spherulites are observed in PEC-g-St.
- 2) The thermal parameters, including melting/crystallization temperature and melting enthalpy, are all as PPH > PEC > IPC. Grafting styrene reduces these parameters except for PEC. Regarding the mechanical parameters, elastic modulus and tensile strength are of PPH > PEC > IPC, while the elongation-at-break is the opposite. Grafting modification tends to deteriorate the mechanical flexibility. However, neither thermal nor mechanical properties are significantly influenced by grafting modification. The elongation-at-break and tensile strength of modified PP can meet the requirement of corresponding IEC standard.
- 3) The dielectric properties are also as PPH > PEC > IPC. For all these PP matrices, grafting styrene can enhance the DC volume resistivity, DC breakdown strength, and suppress the space charge accumulation. PPH-g-St shows the most significant enhancement. The enhancement of PEC is moderate, whereas grafting-modified IPC shows little improvement.
- 4) All the properties mentioned above are strongly correlated with crystallization. IPC shows excellent mechanical properties, but grafting modification is not effective on its dielectric performance. The melting temperature is also at the lowest level. PEC shows moderate enhancement after grafting modification, with a lower melting temperature compared with PPH. Grafting-modified PPH exhibits the best dielectric performance especially under the high temperature of 90 °C, even 110 °C. The thermal properties are also of the top class. Hence, PPH is considered to be the best choice for grafting modification. However, the elastic modulus of PPH is still too high for HVDC cable application and needs further modification.

APPENDIX

The Weibull plots of DC breakdown strength corresponding to Fig. 7 and Table V in Section III-F are shown in Fig. 10.

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